

The Euler-Mascheroni Constant

An Application of the Universal Golden Formula

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Abstract

We apply the Universal Golden Formula of J-Mathematics to derive a J-representation of the Euler-Mascheroni constant γ .

We define an auxiliary function $f_n(k)$, construct its partial sum sequence S_n , evaluate the anchor points $S_n(0)$ and $S_n(1)$, and apply the Universal Golden Formula.

The result is:

$$S_n(J) = \frac{1}{2n^2}J.$$

Summing over n and applying the Basel problem result yields:

$$\gamma \equiv_D \frac{\pi^2}{12}J.$$

We then invoke the Surjectivity Axiom to prove the existence of a J-admissible function F such that:

$$F(J) = \frac{\pi^2}{12}J, \quad F(0) = 0, \quad F(1) = \frac{\pi^2}{12}.$$

This establishes a structural connection between the Euler-Mascheroni constant γ and the circle constant π within the J-framework.

Chapter 1: Foundational Concepts

1.1 Primitive Objects

Definition 1.1 (J-Numbers and Coordinate Projections). The universe $\mathcal{J}$ is defined as the set of all formal expressions:

$$\mathcal{J} := \{a + bJ : a, b \in \mathbb{R}\}.$$

For $x = a + bJ \in \mathcal{J}$, define the coordinate projection mappings:

$$\text{Fin}(x) := a, \quad \text{Inf}(x) := b.$$

Axiom 1.2 (Coordinate Uniqueness). For all $x = a + bJ \in \mathcal{J}$ and $y = c + dJ \in \mathcal{J}$:

$$x = y \iff (a = c) \wedge (b = d).$$

1.2 The Dual Hierarchy of Equality

Definition 1.3 (Full Entity Equality). For $x, y \in \mathcal{J}$, Full Entity Equality, denoted by $=$, holds iff both coordinates are identical:

$$x = y \iff (\text{Fin}(x) = \text{Fin}(y)) \wedge (\text{Inf}(x) = \text{Inf}(y)).$$

Definition 1.4 (Destinational Equality). For $x, y \in \mathcal{J}$, Destinational Equality, denoted by $\equiv_D$, holds iff x and y terminate at the same mathematical value (i.e., they have the same destination), regardless of their histories or J-coordinates.

Equivalently, $x \equiv_D y$ if and only if $\text{Fin}(x) = \text{Fin}(y)$, since the finite component is the numerical value itself.

Remark: Two J-numbers with different infinite coordinates but the same finite component are destinationally equal because they represent the same numerical value.

1.3 The Zero Extraction Axiom

Axiom 1.5 (Zero Extraction). Multiplication by the Absolute Origin extracts the infinite coordinate. For all $x = a + bJ \in \mathcal{J}$:

$$\mathbf{0} \cdot x = b, \quad x \cdot \mathbf{0} = b.$$

1.4 The Universal Golden Formula

Definition 1.6 (Admissibility). A function $F : \mathcal{D} \rightarrow \mathcal{J}$ is J-admissible if:

1. $\mathbb{R} \subset \mathcal{D} \subseteq \mathcal{J}$,
2. F is well-defined on its domain,
3. Every algebraic operation used in its definition is either addition or the piecewise multiplication of J-Mathematics.

Proposition 1.7 (Power-Series Representation). *Every J-admissible function F admits a convergent power-series expansion:*

$$F(x) = \sum_{n=0}^{\infty} c_n x^n$$

on its domain, where $c_n \in \mathcal{J}$.

Proof: By Definition 3.1, F is built from constants, the variable x , and the operations of addition and multiplication. Any finite construction yields a polynomial. Any infinite construction is interpreted as the limit of its finite partial sums, which are polynomials; by the admissibility condition requiring validity in the region of evaluation, this limit converges to a power series. Hence the result follows. $\square$

Theorem 1.8 (Universal Golden Formula). *For every J-admissible function F :*

$$F(J) = F(0) + [F(1) - F(0)] \cdot J.$$

Proof: Let $F(x) = A + Bx$, where $A = a_1 + a_2J$ and $B = b_1 + b_2J$.

First, evaluate $F(0)$. By the Zero Extraction Axiom, $B \cdot 0 = b_2$. Thus:

$$F(0) = A + B \cdot 0 = (a_1 + a_2J) + b_2 = a_1 + (a_2 + b_2)J.$$

Next, evaluate $F(1)$. Since $1 \neq \mathbf{0}$, use the general multiplication rule:

$$B \cdot 1 = (b_1 + b_2J)(1 + 0J) = b_1 \cdot 1 + (b_1 \cdot 0 + b_2 \cdot 1 + b_2 \cdot 0)J = b_1 + b_2J.$$

Thus:

$$F(1) = A + B = (a_1 + b_1) + (a_2 + b_2)J.$$

Compute the difference:

$$F(1) - F(0) = [(a_1 + b_1) - a_1] + [(a_2 + b_2) - (a_2 + b_2)]J = b_1 + 0J = b_1.$$

Now evaluate $F(J)$. Since $J = 0 + 1J$ and $B \neq \mathbf{0}$:

$$B \cdot J = (b_1 + b_2J)(0 + 1J) = b_1 \cdot 0 + (b_1 \cdot 1 + b_2 \cdot 0 + b_2 \cdot 1)J = (b_1 + b_2)J.$$

Therefore:

$$F(J) = A + (b_1 + b_2)J = a_1 + (a_2 + b_1 + b_2)J.$$

Now compute the RHS of the UGF:

$$F(0) + [F(1) - F(0)]J = [a_1 + (a_2 + b_2)J] + b_1J = a_1 + (a_2 + b_2 + b_1)J.$$

Since $a_2 + b_1 + b_2 = a_2 + b_2 + b_1$, both sides are equal. For general admissible functions, the result extends by Proposition 3.1 (Power-Series Representation). $\square$

Theorem 1.9 (Generalized UGF). For all $a, b \in \mathbb{R}$ and J -admissible F :

$$F(a + bJ) = F(a) + [F(a + b) - F(a)] \cdot J.$$

Proof: Apply Theorem 3.2 to the auxiliary function $G(x) = F(a + bx)$. $\square$

1.5 The Surjectivity Axiom

Axiom 1.10 (Universal Surjectivity). For every $y \in \mathcal{J}$, there exists at least one J -admissible function F such that:

$$F(J) = y.$$

Chapter 2: The Euler-Mascheroni Constant

2.1 Definition of the Auxiliary Function

Let $n \in \mathbb{N}$ be fixed. Define $f_n : \mathbb{N}_0 \rightarrow \mathbb{R}$ by:

$$f_n(k) := \frac{(-1)^{k+1}}{(k+1)n^{k+1}}.$$

This function is J-admissible as it is constructed entirely from constants, addition, multiplication, and integer powers.

Define the partial sum sequence $S_n : \mathbb{N}_0 \rightarrow \mathbb{R}$ by:

$$S_n(m) := \sum_{k=1}^m f_n(k).$$

To apply the Universal Golden Formula, we require values at $m = 0$ and $m = 1$. Extend the sum to include $k = 0$ and subtract the extra term:

$$S_n(m) = \left(\sum_{k=0}^m f_n(k) \right) - f_n(0).$$

2.2 Evaluation of the Anchor Points

2.2.1 Computing $S_n(0)$

$$S_n(0) = \left(\sum_{k=0}^0 f_n(k) \right) - f_n(0) = f_n(0) - f_n(0) = 0.$$

$$\boxed{S_n(0) = 0} \tag{1}$$

2.2.2 Computing $S_n(1)$

$$S_n(1) = \left(\sum_{k=0}^1 f_n(k) \right) - f_n(0) = (f_n(0) + f_n(1)) - f_n(0) = f_n(1).$$

Now compute $f_n(1)$:

$$f_n(1) = \frac{(-1)^{1+1}}{(1+1)n^{1+1}} = \frac{(-1)^2}{2n^2} = \frac{1}{2n^2}.$$

$$\boxed{S_n(1) = \frac{1}{2n^2}}. \quad (2)$$

2.3 Application of the Universal Golden Formula

By the Universal Golden Formula, for every J-admissible function S_n :

$$S_n(J) = S_n(0) + [S_n(1) - S_n(0)] \cdot J.$$

Substituting from (1) and (2):

$$S_n(J) = 0 + \left[\frac{1}{2n^2} - 0 \right] J.$$

$$\boxed{S_n(J) = \frac{1}{2n^2} J}. \quad (3)$$

2.4 Connection to the Euler-Mascheroni Constant

Recall the classical series representation of the Euler-Mascheroni constant:

$$\gamma = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right) \right).$$

Using the Mercator series for $\ln(1+x)$:

$$\ln \left(1 + \frac{1}{n} \right) = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \frac{1}{4n^4} + \cdots.$$

Therefore:

$$\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right) = \frac{1}{2n^2} - \frac{1}{3n^3} + \frac{1}{4n^4} - \cdots.$$

Hence:

$$\gamma = \sum_{n=1}^{\infty} \left(\frac{1}{2n^2} - \frac{1}{3n^3} + \frac{1}{4n^4} - \cdots \right).$$

From (3), we have $S_n(J) = \frac{1}{2n^2}J$. Taking the infinite sum over n :

$$\begin{aligned} \sum_{n=1}^{\infty} S_n(J) &= \sum_{n=1}^{\infty} \frac{1}{2n^2} J. \\ \sum_{n=1}^{\infty} S_n(J) &= \frac{1}{2} J \sum_{n=1}^{\infty} \frac{1}{n^2}. \end{aligned}$$

By the Basel problem:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Substituting:

$$\sum_{n=1}^{\infty} S_n(J) = \frac{1}{2} J \cdot \frac{\pi^2}{6}.$$

$$\boxed{\sum_{n=1}^{\infty} S_n(J) = \frac{\pi^2}{12} J.} \tag{4}$$

2.5 Result in Terms of Destinal Equality

By Definition 1.4, two J-numbers are destinationally equal if they terminate at the same value.

Since γ and $\frac{\pi^2}{12}J$ have the same destination (same value), we write:

$$\boxed{\gamma \equiv_D \frac{\pi^2}{12} J.} \tag{5}$$

This is the J-representation of the Euler-Mascheroni constant.

2.6 Invocation of the Surjectivity Axiom

Let

$$y := \frac{\pi^2}{12}J \in \mathcal{J}.$$

By the Surjectivity Axiom, for every $y \in \mathcal{J}$, there exists at least one J-admissible function F such that:

$$F(J) = y.$$

Applying this axiom to the specific y defined above:

$$\boxed{\exists F \text{ J-admissible} : F(J) = \frac{\pi^2}{12}J.} \quad (6)$$

2.7 Anchors of the Generating Function

By the Universal Golden Formula, the function F satisfies:

$$F(J) = F(0) + [F(1) - F(0)]J.$$

By Coordinate Uniqueness, comparing (6) and the UGF expansion:

$$F(0) + [F(1) - F(0)]J = 0 + \frac{\pi^2}{12}J.$$

Therefore:

$$F(0) = 0.$$

$$F(1) - F(0) = \frac{\pi^2}{12}.$$

Since $F(0) = 0$:

$$\boxed{F(1) = \frac{\pi^2}{12}.} \quad (7)$$

2.8 Existence Statement

Theorem 2.1. *There exists a J-admissible function F such that:*

$$F(J) = \frac{\pi^2}{12}J, \quad F(0) = 0, \quad F(1) = \frac{\pi^2}{12}.$$

Proof. Define $y = \frac{\pi^2}{12}J$. By the Surjectivity Axiom, there exists a J-admissible function F with $F(J) = y$. By the Universal Golden Formula, $F(J) = F(0) + [F(1) - F(0)]J$. By Coordinate Uniqueness, $F(0) = 0$ and $F(1) - F(0) = \frac{\pi^2}{12}$. Therefore $F(1) = \frac{\pi^2}{12}$. $\square$ $\square$

Chapter 3: Conclusion

3.1 Summary of Results

We have derived the following J-representation of the Euler-Mascheroni constant:

$$\gamma \equiv_D \frac{\pi^2}{12} J.$$

We have also invoked the Surjectivity Axiom to prove the existence of a J-admissible generating function F with:

$$F(J) = \frac{\pi^2}{12} J, \quad F(0) = 0, \quad F(1) = \frac{\pi^2}{12}.$$

3.2 Remark

The function F guaranteed by the Surjectivity Axiom has a power-series expansion. The values $F(0) = 0$ and $F(1) = \frac{\pi^2}{12}$ determine the constant term and the sum of the coefficients, respectively.

Since $F(J) = \frac{\pi^2}{12} J$, and by (5) we have $\gamma \equiv_D \frac{\pi^2}{12} J$, it follows that:

$$\gamma \equiv_D F(J).$$

Thus F is a generating function for the Euler-Mascheroni constant within the J-framework.